

PLYMOUTH, MA, USA

WMO: 725064

Lat: 41.910N

Lon: 70.729W

Elev: 45

StdP: 100.78

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 54769

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.7	-12.1	-22.9	0.5	-12.1	-20.3	0.6	-9.9	12.4	-1.7	11.3	0.8	3.1	290	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	31.7	22.8	29.8	22.1	28.1	21.1	24.4	29.1	23.6	27.5	22.8	26.2	4.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	17.7	26.3	22.4	17.2	25.7	21.8	16.6	25.1	73.8	29.3	70.5	27.7	67.5	26.2	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.5	8.9	8.0	-18.1	34.7	3.1	1.8	-20.3	36.0	-22.1	37.1	-23.9	38.1	-26.2	39.4		
			-18.9	25.8	3.3	1.0	-21.3	26.5	-23.2	27.0	-25.0	27.6	-27.3	28.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	-1.6	-0.7	2.5	8.0	13.6	18.6	22.2	21.5	17.5	11.6	6.2	1.4	(6)
(7)		DBStd	9.39	5.70	5.29	4.82	4.17	4.17	3.85	3.16	3.19	3.83	4.33	4.75	5.06	(7)
(8)		HDD10.0	1431	361	300	237	87	12	0	0	0	1	32	132	269	(8)
(9)		HDD18.3	3327	619	532	490	310	158	44	5	8	59	214	363	524	(9)
(10)		CDD10.0	1482	1	1	5	28	122	257	379	357	227	81	19	4	(10)
(11)		CDD18.3	335	0	0	0	1	11	51	126	107	36	4	0	0	(11)
(12)		CDH23.3	2554	0	0	1	17	117	412	1010	772	206	19	0	0	(12)
(13)		CDH26.7	738	0	0	0	4	31	121	328	213	40	2	0	0	(13)
(14)	Wind	WSAvg	3.5	3.9	3.9	4.2	3.8	3.5	3.1	3.0	2.7	2.9	3.4	3.5	3.7	(14)
(15)	Precipitation	PrecAvg	1290	116	109	118	111	95	87	82	102	113	110	120	125	(15)
(16)		PrecMax	1784	295	259	346	300	236	275	204	240	338	304	253	286	(16)
(17)		PrecMin	720	32	34	6	29	11	13	23	6	22	18	29	30	(17)
(18)		PrecStd	237	64	52	70	56	49	63	41	54	69	61	62	62	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.0	14.8	21.0	26.0	30.3	32.6	34.0	33.5	30.3	25.9	20.1	17.0	(19)
(20)		MCWB	13.5	12.2	13.7	17.2	20.1	22.9	23.9	24.4	22.6	19.0	16.6	14.7	(20)	
(21)		2%	DB	12.3	12.2	15.1	20.5	26.7	29.8	32.0	30.7	27.5	22.7	17.8	14.1	(21)
(22)		MCWB	10.6	9.5	9.8	13.4	18.3	21.8	23.0	22.9	20.7	18.4	15.2	12.5	(22)	
(23)		5%	DB	9.5	9.8	12.4	17.6	23.6	27.7	30.1	28.9	25.9	20.9	16.1	12.3	(23)
(24)		MCWB	7.6	7.5	9.0	11.8	16.7	20.6	22.4	21.8	20.2	17.3	13.5	10.7	(24)	
(25)		10%	DB	7.0	7.3	10.1	15.2	21.1	25.8	28.3	27.5	23.8	18.8	14.1	9.8	(25)
(26)		MCWB	5.1	4.6	7.0	10.5	15.3	19.1	21.4	21.2	19.7	15.7	11.6	8.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.8	13.2	14.5	17.7	21.7	24.5	25.8	25.5	23.9	21.2	17.9	15.5	(27)
(28)		MCDWB	14.7	14.3	18.1	22.3	28.2	30.7	32.3	31.6	27.8	23.6	19.1	16.7	(28)	
(29)		2%	WB	11.2	10.3	11.9	15.3	19.6	23.0	24.5	24.4	22.7	19.5	16.1	12.9	(29)
(30)		MCDWB	12.0	11.4	14.0	18.9	24.6	28.1	29.6	28.6	25.4	21.7	17.5	14.2	(30)	
(31)		5%	WB	7.9	7.7	9.8	13.5	18.0	21.7	23.6	23.6	21.7	17.9	14.1	10.8	(31)
(32)		MCDWB	9.0	9.1	11.8	15.9	21.9	25.7	27.8	26.9	24.2	19.9	15.7	11.8	(32)	
(33)		10%	WB	5.1	5.0	7.2	11.6	16.4	20.6	22.8	22.8	20.8	16.4	12.2	8.1	(33)
(34)		MCDWB	6.8	7.1	9.2	14.2	19.8	23.9	26.4	25.6	23.2	18.6	13.8	9.3	(34)	
(35)	Mean Daily Temperature Range	MDBR	8.9	9.5	9.4	10.4	10.5	10.3	10.2	10.0	10.1	10.3	9.6	8.9	(35)	
(36)		5% DB	MCDBR	11.5	11.9	12.7	14.6	14.4	13.2	11.9	11.4	11.6	11.7	10.5	10.7	(36)
(37)		MCWBR	10.7	10.0	9.0	8.4	7.4	6.4	5.0	5.0	6.3	7.8	8.4	10.1	(37)	
(38)		5% WB	MCDBR	11.1	11.3	11.6	11.7	12.6	11.2	9.9	9.3	9.1	10.0	9.5	10.4	(38)
(39)		MCWBR	10.9	10.5	9.7	8.2	7.2	6.1	4.8	4.6	5.8	7.9	8.8	10.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.301	0.308	0.331	0.369	0.405	0.456	0.478	0.433	0.372	0.339	0.314	0.301	(40)	
(41)		taud	2.513	2.500	2.481	2.396	2.337	2.232	2.177	2.291	2.442	2.538	2.570	2.565	(41)	
(42)		Ebn at Noon	855	905	919	901	872	825	801	828	862	856	835	827	(42)	
(43)		Edh at Noon	72	85	97	114	124	138	145	125	100	81	68	64	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.77	2.65	3.67	4.73	5.30	5.72	5.95	5.35	4.32	2.92	1.97	1.49	(44)	
(45)		RadStd	0.13	0.25	0.27	0.51	0.54	0.53	0.34	0.26	0.30	0.25	0.14	0.16	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+131	N/A	(46)
(47)	Regional (65 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	+44	(47)

Nomenclature: See separate page